

Spatiotemporal associations between coastal birds and Pacific herring spawning in British Columbia

Full-length Article

[Blinded for review]

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1. Abstract

Pacific herring (*Clupea pallasii*) spawning creates a concentrated coastal resource pulse, but broad-scale bird associations are difficult to estimate from opportunistic observations. We linked eligible submitted complete eBird checklists to event-level Fisheries and Oceans Canada spawn records in registered spatial and temporal windows. Mixed models estimated checklist detection and positive numeric count conditional on detection, with registered effort, observer, temporal, and spatial adjustment. The primary frames contained 217,200 Strait of Georgia and 8,584 West Coast Vancouver Island checklist events. Registered guild and priority-species coefficients varied by region, outcome, and taxon; Surf Scoter and Short-billed Gull coefficients were positive in both primary regions and both response components. Matched 2-km and dominant-observer sensitivities each preserved the matched-reference sign in 15 of 16 components, although the protected sensitivity set retained 43 singular-fit warnings. None of 64 matched whole-bundle placebo components had a Benjamini-Hochberg q-value below 0.05. The prespecified Strait of Georgia specificity panel was non-null for Gadwall (beta = 0.234, 95% CI 0.132–0.337; $q = 7.28 \times 10^{-6}$) and Northern Shoveler (beta = 0.628, 0.527–0.729; $q = 6.56 \times 10^{-34}$). The registered analyses provide confirmatory observational evidence of checklist-conditional associations between recorded local herring-spawn exposure and bird responses. Remaining observation-process limitations and the non-null specificity panel constrain causal and simple ecological-specificity interpretations. The results do not estimate population abundance, biomass, occupancy, migration, or individual movement.

Keywords: Pacific herring; eBird; community science; coastal birds; resource pulse; mixed models; falsification analysis

2. Introduction

Pacific herring spawning deposits eggs and associated organic material in shallow coastal habitats over a brief seasonal interval. Spawn distribution, timing, and recorded magnitude vary substantially among years and coastal areas, while survey coverage and measurement also shape the observed record [1–5]. The government spawn index is a relative index rather than an estimate of absolute biomass, and unavailable components require explicit treatment rather than conversion to zero [6]. These features make spawning a biologically consequential but statistically challenging resource pulse.

British Columbia is a useful setting for examining this pulse because spawning occurs across a long, environmentally heterogeneous coastline and because both birding activity and herring monitoring vary among regions. The same breadth that makes a regional synthesis valuable also prevents a simple substitution of space for replication. Strait of Georgia and West Coast Vancouver Island observations differ in sample size, monitoring geometry, accessibility, and observer composition. We therefore assigned regional inferential roles in advance, treated Strait of Georgia as the primary frame and West Coast Vancouver Island as candidate-primary, and retained the other registered regions as descriptive contexts rather than combining all observations into a coast-wide average.

Localized studies have documented waterbird aggregation, predation on spawn, altered foraging, and movements around herring spawning [7–12]. Responses are not expected to be uniform: birds may use roe, adult herring, carrion, displaced prey, or shoreline resources; timing and accessibility differ among guilds; and some taxa may avoid exposed areas or redistribute outside the observation footprint. Studies at particular sites therefore motivate species- and mechanism-specific expectations but do not, by themselves, establish a coast-wide population response [13].

eBird supplies semistructured complete checklists with observation-process metadata, enabling analyses that distinguish a reported detection from a complete-checklist nondetection and adjust for recorded effort [14,15]. The same open participation that provides broad coverage also creates selection challenges. Observers choose when and where to submit, effort varies, experience differs, and birders may preferentially visit known biological events. Analytical guidance accordingly emphasizes complete-checklist filtering, effort controls, observer structure, spatial and temporal adjustment, validation, and restrained interpretation [16,17]. Such adjustment can reduce measured differences but cannot guarantee exchangeability or eliminate unmeasured site-selection and submission behavior.

To our knowledge, direct integration of complete eBird checklists with event-level DFO Pacific herring spawn information has rarely been reported. We used a frozen Stage 4A design to estimate associations between recorded local herring-spawn exposure and two checklist outcomes: detection and positive numeric count conditional on detection. Our primary estimand is explicitly checklist-conditional. We asked whether registered guild and focal-species outcomes differed with active-near exposure; whether associations

varied among discrete event windows and regions; whether compatible evidence could be partially pooled without mixing estimands or duplicating component evidence; and whether conclusions were stable under matched placebo, spatial-precision, and observer-composition analyses. The registered Stage 4A core is a confirmatory observational analysis; registered secondary analyses remain secondary, and no outcome-informed exploratory claim is promoted to confirmatory status. We foreground the prespecified non-null Strait of Georgia specificity panel because it materially constrains ecological interpretation.

This framing separates statistical confirmation from causal identification. Registration fixes questions, eligible populations, contrasts, and reporting obligations before the relevant response inspection; it does not turn submitted checklists into a probability sample or guarantee that exposure is unconfounded. Accordingly, the confirmatory claims concern associations in the defined checklist population. Secondary analyses evaluate timing and robustness, while falsification and specificity results determine how far ecological interpretation can extend. This distinction allows the completed study to be interpreted on its own terms without treating prospective replication as a prerequisite for the existence of its reported associations.

3. Methods

3.1 Study design and estimand

The analysis used a registered observational design linking complete eBird checklists to recorded Pacific herring-spawn events (Figure 1). The estimand was the adjusted association between recorded active-near exposure and modeled checklist outcomes among eligible submitted complete checklists. It was not a causal effect of spawning. It did not target regional population abundance, biomass, occupancy, migration, individual movement, or total population change.

The design treated the submitted checklist, rather than the bird, observer, or spawn event, as the sampling unit. A detection contrast therefore describes the adjusted difference in the probability that an eligible checklist reported a taxon or guild. The count contrast describes the adjusted difference in log positive numeric count among detected checklists with finite numeric values. These linked but distinct components avoid coding an unquantified X as a numeric count and avoid interpreting a positive-count model as if it included complete-checklist nondetections. Results from one component were not used to fill or reinterpret the other.

Checklist-conditional Stage 4A analysis and evidence audit

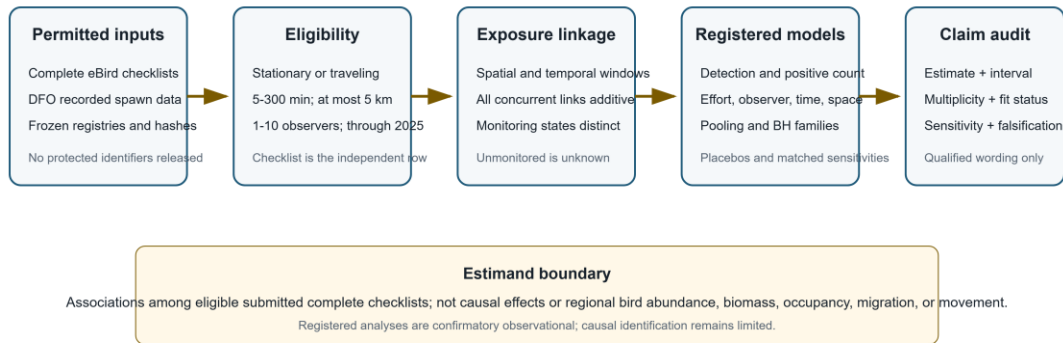


Figure 1. Stage 4A checklist-conditional analysis workflow. The figure summarizes the registered data boundary, complete-checklist eligibility, additive concurrent exposure linkage, model families, and evidence audit. Unmonitored herring coverage remained unknown, and no protected identifiers or locations appear.

3.2 eBird source, eligibility, and response states

The eBird Basic Dataset release was EBD_relMay-2026, but the protected execution was restricted to checklist years through 2025 [18]. Eligible checklists were complete stationary or traveling submissions with duration from 5 to 300 minutes, travel distance no greater than 5 km, one to ten observers, and the registered regional start year. The Strait of Georgia (SoG) frame began in 2005 and was primary; the West Coast Vancouver Island (WCVI) frame began in 2015 and was candidate-primary. Central Coast and North-region frames were hierarchical descriptive contexts.

Complete-checklist omission was converted to a deterministic zero only when the checklist was eligible, the taxon was unreported, and no ambiguity mask made the response structurally unknown. Detection, a positive numeric count, an unquantified X, a lower-bound count, ambiguity, and deterministic omission zero remained distinct. X contributed to detection but never to the numeric-count component. The positive-count response described reported relative count conditional on a finite numeric report greater than zero; it did not estimate absolute abundance or biomass.

Observer count, duration, distance traveled, and protocol were retained as recorded effort covariates. Random intercepts represented observer cluster and generalized location cluster. These terms address measured observation-process heterogeneity; they do not eliminate unmeasured selection. Privacy controls prohibited checklist identifiers, observer identities, exact localities, exact coordinates, event tokens, or transformation mappings from publication artifacts.

3.3 Herring exposure and monitoring states

The exposure source was the official Pacific Herring Spawn Index Data and its associated technical documentation [5,6]. Immutable source points and frozen event metadata were linked to checklists through registered temporal windows and source-point distance strata. The spawn index remained a relative index of spawning biomass; it was never interpreted as absolute biomass.

Monitoring state was explicit. Surveyed-positive, surveyed-negative, and unmonitored-unknown states remained distinct. A missing herring component was not set to zero, and an absent DFO record was not treated as a surveyed negative. Incomplete monitoring can therefore misclassify recorded exposure and is a substantive limitation.

The checklist was the independent analytical row. When a checklist linked concurrently to more than one event, all eligible links contributed additively to six frozen event-time strata and eight frozen distance rings. The analysis did not duplicate the checklist into apparently independent event rows. The primary active-near contrast summarized recorded local exposure; M05 separately represented five discrete windows (immediate pre, spawn start, early egg, late egg, and post). Those windows are registered categories rather than a continuous causal event-study trajectory.

Additive linkage was important because coastal spawning can include overlapping recorded events. Duplicating a checklist once per event would have created artificial replication and changed its statistical weight. Conversely, choosing only one event would have discarded concurrent recorded exposure. The frozen representation retained every eligible link as exposure information within a single analytical row. This addresses bookkeeping for observed concurrent events, but it does not remove interference: biological resources from adjacent events can overlap, recorded source points can imperfectly represent accessible spawn, and an observer's decision to visit a shoreline can respond to the wider event complex.

3.4 Registered outcomes and model families

M01 modeled eight frozen guilds, including a prespecified falsification guild; M02 modeled 49 frozen core taxa. Priority-A species were identified before response inspection. Detection used a binomial-logit model. Positive numeric count used a Gaussian model for log count among positive finite numeric reports. The registered parallel positive-count sensitivity was zero-truncated negative binomial, and all activated model families were reported regardless of sign or apparent interest.

Fixed predictors were event-time stratum, source-point distance stratum, region-period, checklist year, protocol, log duration, log distance traveled plus one, and observer count. Random intercepts represented herring-event block, observer cluster, and generalized location cluster. Publication sensitivity fits used sparse lme4 mixed models [19]: glmer with a binomial-logit likelihood for detection and restricted-maximum-likelihood lmer for

169 log positive count. No simplified row-level generalized linear model replaced the registered
170 mixed architecture.

171 M05 reported discrete event-time coefficients. M29 was a detection-only specificity panel
172 containing Gadwall and Northern Shoveler. The taxa were prespecified as having no
173 verified direct spawn mechanism, but the panel was never defined as a guaranteed
174 biological nonresponder. It tests whether the design also captures shared seasonal,
175 spatial, access, submission, site-selection, or classification structure.

176 3.5 Multiplicity and compatible-family partial pooling

177 Individual coefficients, standard errors, 95% confidence intervals, p-values, and
178 Benjamini-Hochberg (BH) q-values were reported [20]. BH adjustment was confined to
179 coherent model-version, region, and outcome families rather than applied across every
180 exploratory model in the registry. Partial pooling complemented but did not replace
181 individual inference.

182 Historical v1 `partial_pool_*` fields were invalid because some families mixed
183 incompatible model or unit identities and could duplicate component evidence. An initial
184 parser also undercounted the affected scope by treating the registered North-region literal
185 code NA as a missing value. The repair used typed, column-specific parsing, preserved
186 categorical NA, and grouped only compatible model, estimand, response state, unit class,
187 scale, exposure, window, buffer, population, adjustment set, and coefficient meaning. A
188 frozen normal-normal empirical-Bayes method-of-moments estimator produced v2
189 compatible-family posterior summaries. Duplicate M11/M12 representations and
190 noncompleted fits remained explicit NA rather than zero. No v1 artifact was modified, and
191 individual estimates and model-specific BH q-values were exactly preserved.

192 3.6 Matched placebos, sensitivities, and validation

193 M27/M28 v2 moved the complete linked exposure bundle within region-year strata. The
194 bundle included active/reference state, concurrent-link count, all event-time counts, and
195 all distance-ring counts. Deterministic nonzero cyclic shifts had zero fixed points.
196 Transformation construction did not read responses, cross a region or year, duplicate a
197 checklist, or use row-level outcome information. Both placebos used the matched M01
198 mixed architecture.

199 The WCVI 2-km analysis changed only cohort eligibility to the registered high-precision
200 frame. The dominant-observer analysis defined the largest eligible WCVI observer cluster
201 before any guild response was joined and excluded that cluster consistently; no observer
202 token was released. Both retained the matched architecture, so each sensitivity altered
203 one intended dimension. M26 v1 was retired without replacement because its historical
204 summary lacked a registered exposure or visitation contrast.

205 Four deterministic event-blocked folds were the basis for new-event validation. Held-out
206 predictions used fixed effects and set observer and location random effects to their

population expectation; conditional observer or location BLUPs were prohibited. Rows with unsupported fixed factor levels were excluded with aggregate counts, and cells below the release minimum of 20 were suppressed. Observer-disjoint results were interpreted only as observer-composition robustness.

The evidence audit assigned each manuscript claim to its registered estimand, source rows, multiplicity family, relevant sensitivities, and warning state. Confirmatory core findings were retained regardless of sign. Registered timing and sensitivity findings were labeled secondary where appropriate. The specificity panel and model diagnostics were used as interpretation constraints rather than as a post hoc filter that could erase inconvenient primary estimates. This claim-level localization is especially important because a limitation can directly affect one response component or region without logically invalidating all other estimates.

3.7 Singular-fit handling

Singularity was retained as a visible diagnostic rather than an automatic exclusion. For every affected component, the released coefficient, standard error, and interval were finite, but one or more random-effect variance terms were estimated on the boundary. The aggregate diagnostic does not identify which variance term collapsed, so each such fit is interpreted as effectively simpler than the intended three-intercept structure. No headline claim depends exclusively on a singular component. Qualified sensitivity comparisons remain in the main figures, while the full 43-row audit appears in Supplementary Table S6.

3.8 Protected execution and reproducibility

Production response models and protected sensitivities were not rerun for manuscript assembly. The analysis-freeze commit is c54b8e7f95a2fe3573e2e38633079cd223c5a783, tagged stage4a-publication-v2-analysis-freeze. The sensitivity specification and execution code were frozen before protected inputs were read. The execution record reports a maximum response year of 2025 and zero 2026-or-later rows read. Protected inputs are represented by hashes without local paths or row-level content. Manuscript tables and figures were generated only from tracked privacy-safe aggregates.

4. Results

4.1 Analytical sample and execution scope

The frozen analytical frames contained **217,200 eligible checklist events in the Strait of Georgia (SoG; 2005-2025)** and **8,584 on the West Coast of Vancouver Island (WCVI; 2015-2025)**. The hierarchical descriptive frames contained 861 Central Coast and 9,007 North-region events (Table 1). These are checklist-frame sizes, not bird-population counts.

The protected sensitivity execution read no response year later than 2025; the recorded number of 2026-or-later rows read was zero.

Table 1. Frozen analytical populations and inferential roles. The same checklist frame supports multiple taxon outcomes, so these counts must not be summed across taxa.

Region	Period	Eligible checklist events	Inferential role
SoG	2005-2025	217,200	primary
WCVI	2015-2025	8,584	candidate primary
CC	1988-2025	861	hierarchical descriptive
NA	1988-2025	9,007	hierarchical descriptive

4.2 Guild and focal-species associations were heterogeneous

Among the seven ecological guilds in SoG, 5 detection coefficients were positive and 2 were negative; all 7 had BH $q < 0.05$. For positive numeric count conditional on detection, 6 coefficients were positive and 1 negative, with 6 of 7 BH q -values below 0.05. The complete pattern therefore includes both positive and negative associations rather than a universal increase (Figure 2).

WCVI estimates were less uniform. Detection coefficients were positive for 4 of 7 ecological guilds and BH $q < 0.05$ for 3; conditional positive-count coefficients were positive for 6 of 7 guilds and BH $q < 0.05$ for 6. Wider intervals and sparse geometry were most evident in the WCVI component and sensitivity fits.

At the priority-species level, Surf Scoter and Short-billed Gull had positive detection and positive-count coefficients in both primary regions. Glaucous-winged Gull had a positive SoG detection coefficient but an imprecise WCVI detection coefficient, while its conditional count coefficients were positive in both regions. Harlequin Duck and White-winged Scoter were component- or region-dependent (Table 2 and Supplementary Figure S1).

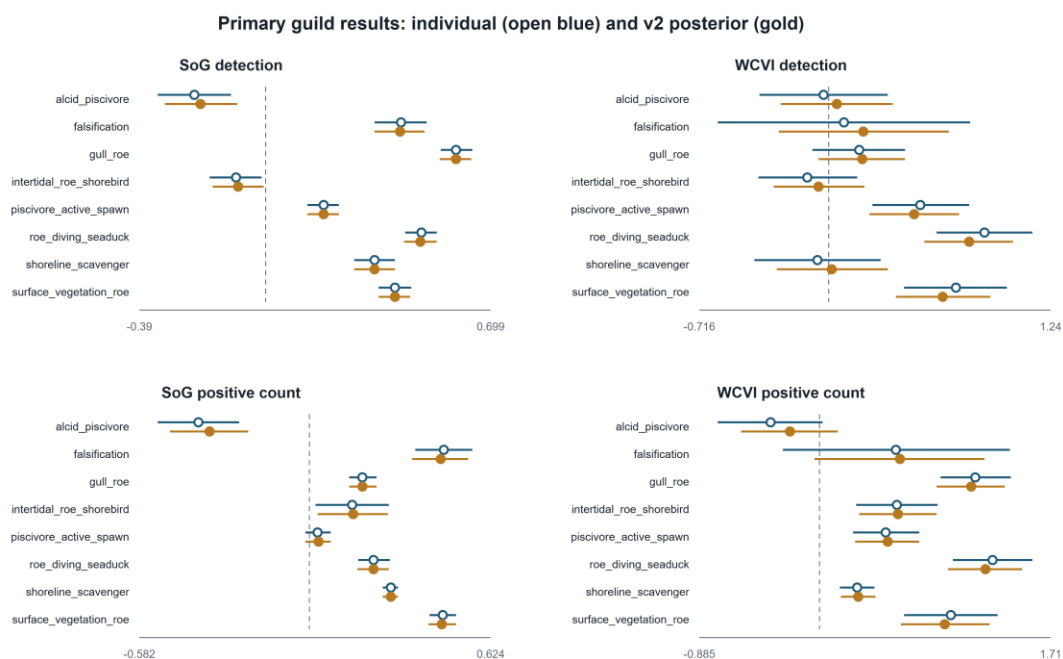


Figure 2. Primary registered guild associations. Coefficients are log odds for checklist detection and log positive count conditional on a numeric detection. Estimates above zero indicate higher modeled outcomes on recorded active-near checklists; they are not causal effects.

Table 2. Priority-A species coefficients. Estimates are log odds for detection or log positive numeric count conditional on detection. BH q-values are model-family adjusted; n is the component checklist count.

Region	Species	Detection	Positive numeric count (detected)
SoG	Glaucous-winged Gull	0.58 (0.53, 0.62); q=1.25e-121; n=217,037	0.22 (0.18, 0.26); q=1.51e-22; n=85,053
SoG	Harlequin Duck	-0.05 (-0.14, 0.05); q=0.423; n=217,200	0.27 (0.18, 0.37); q=6.12e-08; n=12,030
SoG	Short-billed Gull	0.43 (0.37, 0.50); q=3.40e-39; n=217,200	0.39 (0.30, 0.48); q=6.17e-16; n=22,003
SoG	Surf Scoter	0.19 (0.11, 0.26); q=2.62e-06; n=217,183	0.54 (0.42, 0.66); q=8.25e-18; n=16,632
SoG	White-winged Scoter	0.02 (-0.12, 0.15);	0.19 (-0.03, 0.40);

Region	Species	Detection	Positive numeric count (detected)
		q=0.882; n=217,199	q=0.137; n=4,706
WCVI	Glaucous-winged Gull	0.01 (-0.25, 0.26); q=0.976; n=8,568	0.94 (0.69, 1.20); q=7.03e-12; n=4,075
WCVI	Harlequin Duck	0.50 (0.00, 1.00); q=0.070; n=8,584	0.20 (-0.19, 0.58); q=0.455; n=397
WCVI	Short-billed Gull	1.04 (0.74, 1.33); q=4.58e-11; n=8,584	1.32 (0.99, 1.66); q=3.76e-13; n=1,421
WCVI	Surf Scoter	0.97 (0.68, 1.26); q=3.45e-10; n=8,581	1.17 (0.79, 1.55); q=1.89e-08; n=1,857
WCVI	White-winged Scoter	0.70 (0.29, 1.11); q=0.002; n=8,584	0.42 (-0.04, 0.88); q=0.157; n=701

4.3 Discrete event-time patterns differed among regions and outcomes

The 160 registered M05 rows showed window-, guild-, outcome-, and region-specific coefficients rather than a single monotone event trajectory (Figure 3). In SoG, the immediate-pre window had BH $q < 0.05$ for seven of eight guilds in each outcome, while the post window had seven of eight adjusted associations in each outcome and a negative median coefficient. WCVI patterns differed: detection associations were most frequent in the immediate-pre and post windows, whereas positive-count contrasts were more variable. These discrete coefficients are not a continuous causal event study.

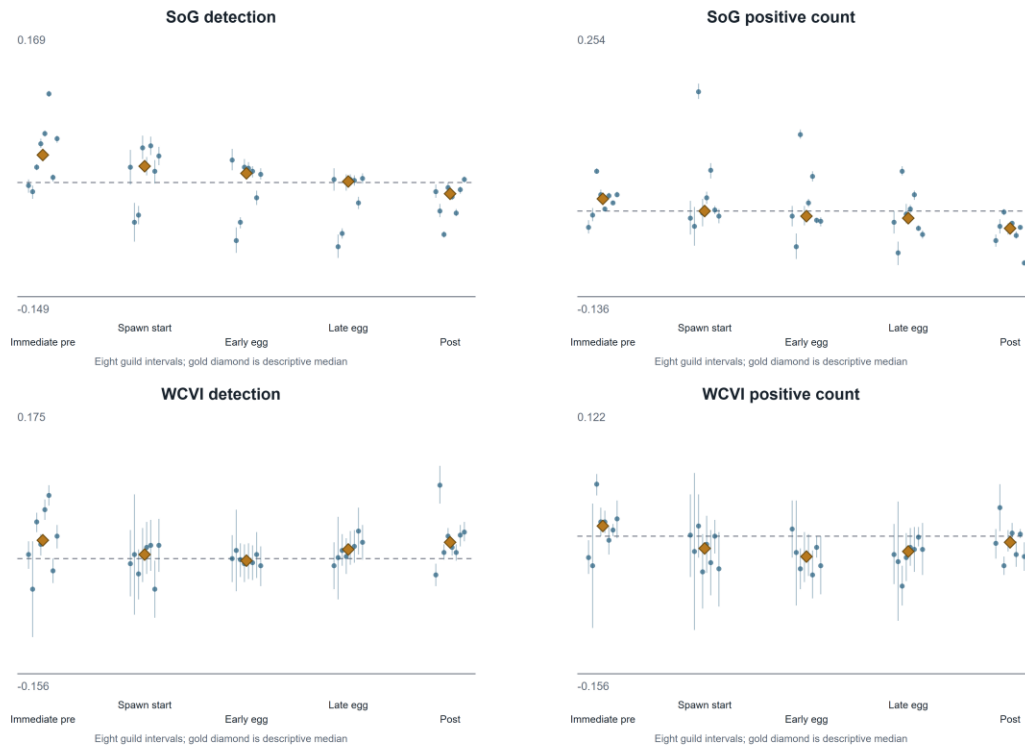


Figure 3. Registered discrete event-time coefficients. Five frozen windows were modeled for each guild, region, and hurdle component. The plot displays uncertainty and includes the falsification guild; the median is descriptive and not a new pooled estimand.

4.4 Matched sensitivities generally preserved direction but included boundary fits

All **128 of 128** protected sensitivity components completed: 85 ordinary completions and 43 completions with an explicit singular-fit warning; none failed. In each of the WCVI 2-km and dominant-observer comparisons, 15 of 16 components retained the sign of the matched sparse M01 reference (Figure 4). Because 23 sensitivity/reference components shown in Figure 4 were singular, this is qualified directional robustness rather than evidence that the intended random-effects structure was fully supported.

The M27/M28 whole-bundle diagnostics completed all 64 components, and none had BH $q < 0.05$ (Figure 5). These null shifted-exposure diagnostics do not establish exchangeability or causal identification.

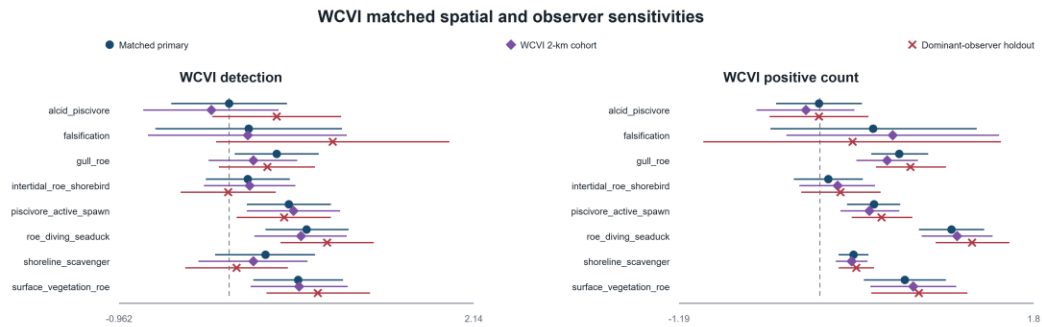


Figure 4. WCVI matched spatial and observer sensitivities. Each sensitivity changed one registered dimension while retaining the sparse mixed-model architecture. Singular fits are retained and disclosed, so visual agreement is qualified robustness rather than proof of a fully supported covariance structure.

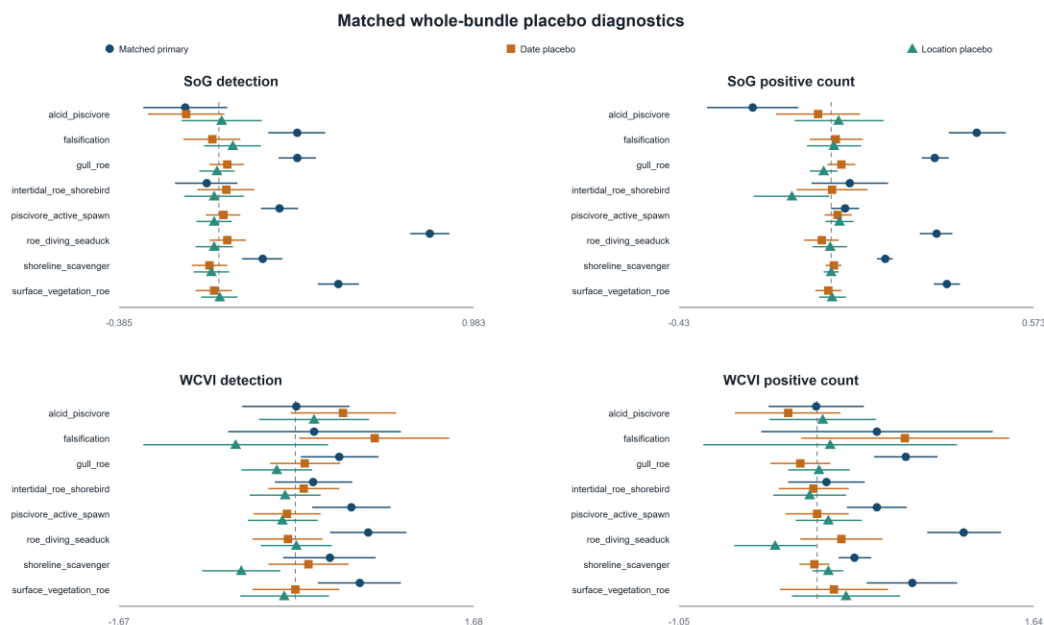


Figure 5. Matched whole-bundle placebo diagnostics. M27 and M28 shifted complete exposure bundles within region-year strata without reading responses. No placebo component had BH q below 0.05; null diagnostics do not establish causal identification.

4.5 The Strait of Georgia specificity panel was non-null

The prespecified SoG specificity panel was non-null. Gadwall detection was positively associated with recorded active-near exposure (log-odds coefficient 0.234, 95% CI 0.132 to 0.337; BH q = 7.28e-06), as was Northern Shoveler detection (0.628, 0.527 to 0.729; BH q = 6.56e-34). Both used the true exposure rather than a shifted placebo, and their magnitudes were within the range of primary SoG guild-detection coefficients. The panel therefore identifies plausible shared seasonal, spatial-access, checklist-submission, site-selection, or exposure-classification structure. It weakens causal and simple species-specific ecological interpretations of SoG detection associations, but it does not show that every association is spurious and it does not test the conditional positive-count component.

The prespecified Strait of Georgia specificity panel was non-null for Gadwall and Northern Shoveler. This indicates that recorded active-near exposure captured shared seasonal, spatial-access, checklist-submission, site-selection, or exposure-classification structure in addition to any taxon-specific ecological response. The panel therefore limits causal and simple species-specific interpretations, but does not negate the checklist-conditional associations or directly test conditional positive-count responses.

Table 3. Sensitivity, placebo, singular-fit, and specificity-panel summary. Component-level estimates and explicit warning states are supplied in the companion CSV tables. Three headline claims include some singular-warning support, but each also has ordinary-

322 fit or otherwise stable evidence; no headline claim depends exclusively on a singular
323 component.

Analysis	Components	Result	Singular warnings	Interpretation
M27/M28 whole-bundle placebos	64	0/64 BH $q < 0.05$	20	diagnostic; not proof of causal identification
WCVI 2-km matched sensitivity	16	15/16 same sign as matched reference	8	qualified spatial- precision robustness
WCVI dominant- observer holdout	16	15/16 same sign as matched reference	9	qualified observer- composition robustness
SoG M29 specificity panel	2	2/2 detection coefficients BH $q < 0.05$	0	material warning against simple ecological specificity

324 4.6 Pooling repair preserved individual inference and made exclusions 325 explicit

326 The authoritative pooling repair invalidated 6,562 finite v1 partial-pooling rows in 112
327 historical families. Typed, column-specific parsing preserved the registered North-region
328 literal code NA, and v2 produced 162 compatible estimable families. The release contains
329 6,085 posterior rows, with 439 duplicate M11/M12 representations and 38 noncompleted
330 rows retained as explicit NA. Individual estimates, standard errors, intervals, p-values, and
331 model-specific BH q-values were preserved; v1 artifacts were not modified.

332 The complete 162-family table, 6,562-row disposition, duplicate-resolution crosswalk,
333 reason codes, and artifact hashes are supplied as supplementary machine-readable
334 tables. The repair changes only the compatible-family synthesis and its publication
335 representation.

5. Discussion

5.1 Associations were consistent with a short local resource pulse, but not uniformly

The registered models identified substantial heterogeneity. Several roe-feeding, surface-feeding, and active-spawn guilds had positive checklist-detection or conditional-count coefficients, and Surf Scoter and Short-billed Gull showed positive coefficients across both primary regions and both hurdle components. These patterns are ecologically consistent with localized aggregation or increased checklist reporting around spawning, in line with site-based studies of waterbird distribution, roe use, and scoter behavior [8–11]. The study does not show that every bird group responds positively. Negative, imprecise, component-dependent, and region-dependent coefficients are part of the registered result.

Detection and conditional count address different processes. A detection coefficient can change when the chance of reporting at least one bird changes, whereas the positive-count component describes the reported relative count after a positive finite numeric observation. The latter excludes X and cannot be translated into biomass or total bird abundance. Divergence between components is therefore scientifically meaningful rather than an inconsistency to be removed.

The cross-region patterns for Surf Scoter and Short-billed Gull are among the clearest focal-species findings because positive estimates occurred in both hurdle components in both primary frames. Even these results remain associations among submitted checklists. They are consistent with local concentration, redistribution within observable habitat, increased detectability, or changes in where and when observers submitted checklists. For other priority species, regional or component dependence is itself informative: it argues against a universal response and is compatible with differences in diet, shoreline access, migration timing, flocking behavior, and exposure measurement. The registered negative and imprecise estimates define the limits of the resource-pulse pattern rather than serving as results to be screened away.

Timing also resisted a single narrative. SoG and WCVI differed in which discrete windows had the most frequent adjusted associations, and post-window medians could be negative even when some guild coefficients were positive. These patterns could reflect taxon-specific access to roe, adult herring, or displaced prey; migration phenology; delayed bird or birder visitation; and error in assigning recorded event dates. They do not identify a continuous temporal response or causal lead-lag process.

5.2 Regional and sensitivity evidence requires qualification

WCVI coefficients were often wider than SoG coefficients because the frame was smaller and observer/location geometry was sparse. The 2-km and dominant-observer analyses each preserved the matched-reference sign in 15 of 16 components, which argues against

wholesale dependence on the broader spatial frame or one dominant observer cluster. However, many WCVI detection fits were singular. A boundary estimate means that the fitted covariance structure effectively collapsed toward a simpler model, even though fixed-effect coefficients and intervals remained finite. These results support directional stability with qualification, not a claim that observer composition or spatial precision is irrelevant.

The sensitivity evidence is most informative when read component by component. Sign preservation shows that the direction of most matched WCVI contrasts did not depend on either the broader eligibility radius or the dominant observer cluster, but it does not imply equal precision, magnitude, or random-effects support. A singular component remains useful when its finite coefficient is interpreted alongside ordinary-fit evidence and the warning is disclosed; it should not become the uniquely decisive basis for a headline claim. This is why three headline claims retain some singular-warning support while none relies exclusively on it. The ordinary-fit and otherwise stable components justify retaining those claims with qualification instead of automatically relabeling the entire result exploratory.

Event-blocked validation retained all four registered folds and excluded unsupported factor levels transparently. Validation describes performance on held-out event blocks under the frozen predictor structure. It does not repair unmeasured site selection, transform community-science submissions into a probability sample, or establish causal transportability.

5.3 The non-null specificity panel localizes an important threat to interpretation

The SoG M29 panel is not interchangeable with the matched shifted-exposure placebos. M27/M28 ask whether response-blind, within-region-year reassignments of the complete exposure bundle reproduce the matched primary pattern; none of their 64 components had BH q below 0.05. M29 instead uses the true active-near exposure for two prespecified taxa without a verified direct herring-spawn mechanism. Both M29 coefficients were positive and adjusted-significant.

Because the M29 magnitudes overlap the ecological guild-detection range, the panel materially weakens a simple interpretation in which positive SoG detection coefficients arise only from direct spawn-resource use. Plausible explanations include residual seasonality, shoreline habitat and access, birder site choice, post-exposure visitation, submission behavior, and exposure classification. The panel does not prove that every primary association is spurious: it is detection-only, does not target the conditional positive-count component, and does not reproduce every species, guild, region, or window. The defensible conclusion is narrower: SoG detection associations are real features of the eligible submitted-checklist data, but their ecological specificity and causal interpretation are limited.

Independent replication and structured local monitoring would help distinguish ecological response from residual observation-process and exposure-classification mechanisms.

5.4 Pooling and complete-family reporting reduce selective interpretation

The pooling repair matters because partial pooling can appear precise while silently mixing incompatible targets or counting the same implementation component twice. Typed identities and explicit duplicate dispositions restored one coherent evidence unit per compatible family. Keeping all individual estimates visible prevents the posterior summaries from masking heterogeneity. Reporting the complete registered families, including negative, null, failed, singular, and falsification results, further reduces the temptation to organize conclusions around significance alone.

Several features strengthen the observational evidence. The response states were defined from complete-checklist semantics rather than treating every record as an interchangeable count; exposure timing and distance were registered; concurrent links retained a single checklist weight; and the primary, sensitivity, placebo, and specificity results used an explicit evidence matrix. The pooling repair preserved every individual estimate and model-specific BH q-value while preventing incompatible or duplicated evidence from acquiring synthetic precision. Together, these controls support reproducibility and make the scope of statistical confirmation transparent. Their value is not that they eliminate selection bias, but that they prevent avoidable analytical ambiguities from being confused with the remaining ecological and observation-process uncertainty.

6. Limitations

First, checklist submission and site selection are not random. Observers may visit accessible shorelines or known spawn sites, and visitation may change after public knowledge of an event. Recorded effort variables and observer/location structure reduce measured differences but cannot eliminate unmeasured selection. The non-null SoG specificity panel demonstrates that observation-process or shared environmental structure remains plausible.

Second, recorded spawn exposure is imperfect. DFO monitoring varies in space and time; unmonitored areas remain unknown rather than negative. Recorded dates, source points, and index components may not coincide perfectly with biological availability to birds. Concurrent events can interfere, although all registered concurrent links were retained additively rather than duplicating checklists.

Third, WCVI has sparse geometry and concentrated observer participation. Matched 2-km and dominant-observer results are useful, but singular-fit warnings show that some intended random-effect variance components were weakly identified. The released

diagnostic cannot identify the exact collapsed variance term, preventing a more granular interpretation of those boundary fits.

Fourth, the two-part response is conditional on eBird reporting. A complete-checklist nondetection is not proof of biological absence; a positive numeric count is not a census; X is detection-only; and ambiguity can create structural unknowns. The models do not estimate occupancy or detection probability in a formal repeated-survey occupancy design.

Fifth, the registered design supports confirmatory observational inference about checklist-conditional associations, but it cannot by itself distinguish ecological response from every observation-process or exposure-classification mechanism. This limits causal and simple ecological-specificity interpretation without negating the observed statistical associations. We did not access 2026-or-later responses, initiate Stage 4B, or add outcome-driven variants for manuscript preparation.

These limitations identify specific interpretive boundaries. Selection affects generalization beyond eligible submitted checklists; incomplete monitoring and timing error affect the meaning of recorded exposure; sparse WCVI geometry affects precision and support for the specified random-effects structure; and the two-part reporting process prevents population-level interpretation. None changes the fact that the frozen models estimate the reported checklist-conditional contrasts. A useful next step would pair repeatable local bird surveys with explicit records of shoreline access, observer recruitment, visitation before and after public spawn reports, and contemporaneous measures of spawn availability. Such monitoring could separate biological use of the resource pulse from changes in detectability and submission behavior while testing whether the regional patterns recur.

7. Conclusions

The registered analyses provide confirmatory observational evidence of checklist-conditional associations between recorded local herring-spawn exposure and bird responses. Associations varied among guilds, focal species, regions, time windows, and response components. Matched WCVI sensitivities generally preserved direction and matched whole-bundle placebos were not BH-significant, but boundary fits and the non-null SoG specificity panel require prominent qualification. Remaining observation-process limitations and the non-null specificity panel constrain causal and simple ecological-specificity interpretations; they do not negate the reported associations. The study does not identify a causal effect or a change in regional bird abundance, biomass, occupancy, migration, movement, or total populations.

8. Data availability

DFO-derived source data are available through the Government of Canada Open Government Portal [6], subject to the source record and its monitoring caveats. eBird checklist data require authorized access under eBird terms and are not redistributed [18]. Raw EBD/SED, protected row-level derivatives, observer identities, exact localities, exact coordinates, event tokens, and transformation mappings cannot be shared because of source terms and privacy controls. Privacy-safe aggregate tables, claim audits, provenance records, and publication figures are deposited in the project repository [repository URL withheld in this optional blinded companion]. An authorized researcher can reconstruct the analysis from permitted source data, frozen registries, and the tagged code.

9. Code availability

Code, frozen specifications, privacy-safe aggregate outputs, and reproducibility instructions are available in the project repository [repository URL withheld in this optional blinded companion]. The exact analysis freeze identifier will be restored in the accepted unblinded version. Manuscript assembly did not refit response models or alter any frozen v1 or v2 analysis artifact.

10. Declarations

CRedit authorship contribution statement: Blinded for review.

Funding: Blinded for review.

Declaration of competing interest: Blinded for review.

Ethics and permits: Not applicable; the study used secondary observational datasets and involved no animal handling or human-participant research.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process: Blinded for review; the complete proposed declaration is retained in the unblinded manuscript.

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12. References

- [1] Haegele CW, Schweigert JF. Distribution and characteristics of herring spawning grounds and description of spawning behavior. *Can J Fish Aquat Sci* 1985;42:s39–55. <https://doi.org/10.1139/f85-261>.
- [2] Hay DE, Kronlund AR. Factors affecting the distribution, abundance, and measurement of pacific herring spawn. *Can J Fish Aquat Sci* 1987;44:1181–94. <https://doi.org/10.1139/f87-141>.
- [3] Hay DE, McCarter PB, Daniel KS, Schweigert JF. Spatial diversity of pacific herring spawning areas. *ICES J Mar Sci* 2009;66:1662–6. <https://doi.org/10.1093/icesjms/fsp139>.
- [4] Rooper CN, Boldt JL, Cleary J, Pena AM, Thompson M, Grinnell M. Evaluating factors affecting the distribution and timing of pacific herring spawn in british columbia. *Mar Ecol Prog Ser* 2024;741:251–69. <https://doi.org/10.3354/meps14274>.
- [5] Grinnell MH, Schweigert JF, Thompson M, Hawkshaw S, Cleary JS. [Calculating the spawn index for pacific herring \(clupea pallasii\) in british columbia, canada](#). Fisheries; Oceans Canada; 2023.
- [6] Fisheries and Oceans Canada. [Pacific herring spawn index data](#) 2026.
- [7] Haegele CW. Seabird predation of pacific herring, clupea pallasii, spawn in british columbia. *Can Field-Nat* 1993;107:73–82. <https://doi.org/10.5962/p.357075>.
- [8] Sullivan TM, Butler RW, Boyd WS. Seasonal distribution of waterbirds in relation to spawning pacific herring, clupea pallasii, in the strait of georgia, british columbia. *Can Field-Nat* 2002;116:366–70. <https://doi.org/10.5962/p.363475>.
- [9] Rodway MS, Regehr HM, Ashley J, Clarkson PV, Goudie RI, Hay DE, et al. Aggregative response of harlequin ducks to herring spawning in the strait of georgia, british columbia. *Can J Zool* 2003;81:504–14. <https://doi.org/10.1139/z03-032>.
- [10] Lewis TL, Esler D, Boyd WS. Foraging behaviors of surf scoters and white-winged scoters during spawning of pacific herring. *Condor* 2007;109:216–22. <https://doi.org/10.1093/condor/109.1.216>.
- [11] Lok EK, Kirk M, Esler D, Boyd WS. Movements of pre-migratory surf and white-winged scoters in response to pacific herring spawn. *Waterbirds* 2008;31:385–93. <https://doi.org/10.1675/1524-4695-31.3.385>.
- [12] Lok EK, Esler D, Takekawa JY, De La Cruz SEW, Boyd WS, Nysewander DR, et al. Spatiotemporal associations between pacific herring spawn and surf scoter spring migration: Evaluating a silver wave hypothesis. *Mar Ecol Prog Ser* 2012;457:139–50. <https://doi.org/10.3354/meps09692>.

- 547 [13] Kelly JP, Rothenbach CA, Weathers WW. Echoes of numerical dependence:
548 Responses of wintering waterbirds to pacific herring spawns. *Mar Ecol Prog Ser*
549 2018;597:243–57. <https://doi.org/10.3354/meps12594>.
- 550 [14] Sullivan BL, Wood CL, Iliff MJ, Bonney RE, Fink D, Kelling S. eBird: A citizen-based
551 bird observation network in the biological sciences. *Biol Conserv* 2009;142:2282–92.
552 <https://doi.org/10.1016/j.biocon.2009.05.006>.
- 553 [15] Kelling S, Johnston A, Bonn A, Fink D, Ruiz-Gutierrez V, Bonney R, et al. Using
554 semistructured surveys to improve citizen science data for monitoring biodiversity.
555 *BioScience* 2019;69:170–9. <https://doi.org/10.1093/biosci/biz010>.
- 556 [16] Johnston A, Fink D, Hochachka WM, Kelling S. Estimates of observer expertise
557 improve species distributions from citizen science data. *Methods Ecol Evol* 2018;9:88–97.
558 <https://doi.org/10.1111/2041-210X.12838>.
- 559 [17] Johnston A, Hochachka WM, Strimas-Mackey ME, Ruiz Gutierrez V, Robinson OJ,
560 Miller ET, et al. Analytical guidelines to increase the value of community science data: An
561 example using eBird data to estimate species distributions. *Divers Distrib* 2021;27:1265–
562 77. <https://doi.org/10.1111/ddi.13271>.
- 563 [18] eBird. [eBird basic dataset, version EBD_relMay-2026](#) 2026.
- 564 [19] Bates D, Maechler M, Bolker B, Walker S. Fitting linear mixed-effects models using
565 lme4. *J Stat Softw* 2015;67:1–48. <https://doi.org/10.18637/jss.v067.i01>.
- 566 [20] Benjamini Y, Hochberg Y. Controlling the false discovery rate: A practical and
567 powerful approach to multiple testing. *J R Stat Soc Ser B Stat Methodol* 1995;57:289–300.
568 <https://doi.org/10.1111/j.2517-6161.1995.tb02031.x>.